

git reset, revert and tag commands

git reset is used to move or remove commits

Syntax.

git reset [--soft | --mixed | --hard] HEAD^

<u>Type</u>	<u>Effect on staging</u>	<u>Effect on working Directory</u>	<u>Effect on commit history</u>
soft	keeps staged	keeps changes	Removes commit only
mixed	clears staging	keeps changes	removes commit
hard	clears everything	Discards all changes	removes commit and code

Example :-

git reset --soft HEAD^

Only removes the last commit, but your code is still staged

`git reset --mixed HEAD~1` #or just `git reset HEAD~1`

Remove the commit and unstages the code.

`git reset --hard HEAD~1`

completely remove the commit and also delete the code changes

git revert - Safely undo a commit
(Recommended for shared branches)

command

`git revert <commitID>`

- This creates a new commit that undoes the changes of the given commit.

- A popup will open for commit message, where you can edit and save.

- Safe to use because history stays intact.

git tag :- create markers (like "v1.0") for releases

1. Lightweight tag (just a pointer)

`git tag v1.0 commitID`

- NO message, just marks the commit.

2. Annotated tag (Heavy)

`git tag v1.0 commitID -m "Initial release version"`

- includes message, author info, and date

Other tag commands

Task

List all tags

command

git tag

view details of a tag

git show tagname

Delete a tag locally

git tag -d tagname

push all tags to github

git push origin master
--tags

Summary Table of above all commands.

command

Description

git reset HEAD~1

Undo last commit (default =
mixed)

git reset --soft HEAD~1

Keep staged code, remove
commit

git reset --hard HEAD~1

Remove commit and code
completely

git revert commitID

Undo a commit by creating a
reverse commit

git tag tagname commitID
-m " "

create annotated (heavy) tag

git tag

List all tags

git show tagname

show tag details

git tag -d tagname

Delete tag locally

git push origin master
--tags

push all tags to remote